Statistical tests used in Ex2

Pearson χ^2, Wilks' theorem, and the per-bin LRT

Supplementary material to Computational Genomics, Exercise 2 (Library Complexity)

Roee Tekoah
roee.tekoah@mail.huji.ac.il

Scope

A self-contained primer for the three tests used in Ex2: Pearson's chi-squared goodness-of-fit (Q2(b)), Wilks' correction for parameters fit from the data (Q2(b) and Q3(b)), and the likelihood-ratio test (LRT) for shared λ across files (Q3(b)). The three rest on the same large-sample machinery; we develop it once and apply it to each.

Contents

1 Pearson's χ^2 goodness-of-fit **2**
1.1 The setting and the statistic . . . 2
1.2 Multinomial recap: the count vector and its covariance . . . 2
1.3 Why the denominator is exp[k]: variance, Poisson limit, z-score . . . 3
1.4 The null distribution: $\chi^2(K-1-p)$ . . . 4
1.5 Cochran's rule – when the Gaussian approximation breaks . . . 5
1.6 From the general framework to Q2(b): two roles of "Poisson" . . . 6

2 Wilks' correction for estimated parameters **6**
2.1 What changes when the model's parameters are fit from the data . . . 6
2.2 The geometric picture . . . 7
2.3 Wilks' theorem (1938) . . . 7
2.4 Practical effect in Q2(b): why χ^2 is in the tens of thousands . . . 7

3 Likelihood-ratio test (LRT) **8**
3.1 What an LRT is, in one paragraph . . . 8
3.2 Null distribution (Wilks) . . . 8

4 Q3(b) LRT in detail **9**
4.1 Exactly what we are testing . . . 9
4.2 The log-likelihoods . . . 9
4.3 Computing the two MLEs . . . 10
4.4 The test statistic and decision rule . . . 10
4.5 Practical considerations . . . 11

5 Reading the Q2(b) and Q3(b) results **11**
5.1 Q2(b) . . . 11
5.2 Q3(b) . . . 11
5.3 Translating back to the science . . . 11

1 Pearson's χ^2 goodness-of-fit

1.1 The setting and the statistic

We have an empirical histogram on K cells, with counts $\text{obs}[k]$ for $k = 1, \ldots, K$ summing to U total observations. A parametric model with parameter θ assigns each cell a probability $\pi_k(\theta)$ with $\sum_k \pi_k(\theta) = 1$. The model-predicted count in cell k is

$$\exp[k] \;=\; U\,\pi_k(\hat{\theta}),$$

where $\hat{\theta}$ is an estimator of θ (typically MLE).

The question is: could the observed counts have come from the model, up to sampling noise? Pearson's test statistic is

$$\chi^2_{\text{Pearson}} \;=\; \sum_{k=1}^{K} \frac{\big(\text{obs}[k] - \exp[k]\big)^2}{\exp[k]}.$$

Two natural questions: why is the denominator $\exp[k]$, and what distribution does the statistic follow under the null hypothesis that the model is correct? Answering them takes the next three subsections.

1.2 Multinomial recap: the count vector and its covariance

If you remember the multinomial distribution cold, skim this. Otherwise, the right starting point is to remember that the counts vector is multinomial and understand its covariance structure – both the denominator $\exp[k]$ and the eventual degrees-of-freedom $K - 1 - p$ come from there.

Binomial first. $X \sim \text{Binomial}(U, p)$ counts successes in U independent yes/no trials with success probability p. Then $\mathbb{E}[X] = Up$ and $\text{Var}(X) = Up(1-p)$.

Multinomial: K categories at once. Each trial now picks one of K categories with probabilities $\pi_1, \ldots, \pi_K$ (summing to 1). The count vector $(X_1, \ldots, X_K) \sim \text{Multinomial}(U;\, \pi_1, \ldots, \pi_K)$ records how many trials fell in each. Means:

$$\mathbb{E}[X_k] \;=\; U\pi_k \;=\; \exp[k].$$

Each individual coordinate is binomial – for any one trial, it either lands in category k or it doesn't – so $X_k \sim \text{Binomial}(U, \pi_k)$ and

$$\text{Var}(X_k) \;=\; U\pi_k(1-\pi_k).$$

This already explains the variance bit of the Pearson denominator: in the "Poisson regime" where π_k is small, $(1-\pi_k) \approx 1$ and the variance simplifies to $U\pi_k = \exp[k]$. So $\sqrt{\exp[k]}$ is the standard deviation of a typical cell count, and $\sqrt{\exp[k]}$ is the natural scale to measure deviations against. We use this below.

The bins are not independent. The constraint $X_1 + X_2 + \cdots + X_K = U$ holds exactly: each trial is counted by exactly one X_k. If one bin happens to overshoot, the others must undershoot to keep the sum equal to U. Concretely the counts are negatively correlated:

$$\text{Cov}(X_k, X_\ell) \;=\; \begin{cases} U\pi_k(1-\pi_k) & k = \ell \\ -U\pi_k\pi_\ell & k \neq \ell \end{cases}.$$

A quick numeric example: $U = 100$, $\pi = (0.2, 0.5, 0.3)$ gives expected counts $(20, 50, 30)$, diagonal variances $(16, 25, 21)$, and off-diagonal covariances $-10, -6, -15$. The covariance matrix is

$$\Sigma \;=\; \begin{pmatrix} 16 & -10 & -6 \\ -10 & 25 & -15 \\ -6 & -15 & 21 \end{pmatrix},$$

and *every row sums to zero*: $16 - 10 - 6 = 0$, $-10 + 25 - 15 = 0$, $-6 - 15 + 21 = 0$. That is not a coincidence – Σ is singular with the all-ones vector in its null space, because there is no randomness in the direction that moves the total U (the total is fixed by definition). The deviation vector $d_k = \text{obs}[k] - \exp[k]$ inherits the same constraint: $\sum_k d_k = 0$. We keep this in mind for the degrees-of-freedom argument later.

1.3 Why the denominator is $\exp[k]$: variance, Poisson limit, z-score

A single per-cell term in Pearson's statistic, written as

$$\frac{(\text{obs}[k] - \exp[k])^2}{\exp[k]} \;=\; \left(\frac{\text{obs}[k] - \exp[k]}{\sqrt{\exp[k]}}\right)^2 \;=\; z_k^2,$$

is just a squared z-score: how many standard deviations $\text{obs}[k]$ is from its expected value. Three pieces back this up:

(1) Variance of $\text{obs}[k]$ from Bernoulli arithmetic. Decompose $\text{obs}[k]$ as a sum of indicators: define $X_i = 1$ if marble i lands in cell k, 0 otherwise, so each $X_i \sim \text{Bernoulli}(\pi_k)$ and $\text{obs}[k] = X_1 + \cdots + X_U$. Since $X_i \in \{0, 1\}$ we have $X_i^2 = X_i$, hence

$$\mathbb{E}[X_i] = \pi_k, \qquad \text{Var}(X_i) = \mathbb{E}[X_i^2] - \mathbb{E}[X_i]^2 = \pi_k - \pi_k^2 = \pi_k(1 - \pi_k).$$

Independence gives $\text{Var}(\text{obs}[k]) = \sum_i \text{Var}(X_i) = U\pi_k(1 - \pi_k)$. (Intuition: if $\pi_k \in \{0, 1\}$ there's no randomness; variance peaks at $\pi_k = 1/2$ where each marble is most uncertain.)

(2) Poisson regime: variance $\approx$ mean. When π_k is small (each cell gets a small share of the marbles), $1 - \pi_k \approx 1$, so $\text{Var}(\text{obs}[k]) \approx U\pi_k = \exp[k]$. This is the Poisson limit: as $U \to \infty$ and $\pi_k \to 0$ with $U\pi_k = \lambda$ fixed, $\text{Binomial}(U, \pi_k) \to \text{Poisson}(\lambda)$, and Poisson distributions always satisfy mean = variance.

(3) Standardising gives $N(0, 1)$. The CLT says a sum of many independent small effects looks Gaussian. So for U large,

$$\text{obs}[k] \;\approx\; N\big(\exp[k],\, \exp[k]\big).$$

Standardising any $N(\mu, \sigma^2)$ produces a standard normal: if $X \sim N(\mu, \sigma^2)$, then $Z = (X - \mu)/\sigma \sim N(0, 1)$. Plugging $\mu = \exp[k]$ and $\sigma = \sqrt{\exp[k]}$:

$$z_k \;=\; \frac{\text{obs}[k] - \exp[k]}{\sqrt{\exp[k]}} \;\approx\; N(0, 1).$$

So each Pearson per-cell term is exactly the squared z-score of one bin, and the full statistic is $\sum_k z_k^2$.

1.4 The null distribution: $\chi^2(K-1-p)$

We now know $\chi^2_{\text{Pearson}} = \sum_k z_k^2$ with each $z_k \approx N(0,1)$ under H_0. The remaining question is which χ^2 distribution that sum follows.

What $\chi^2(d)$ even means. The chi-squared distribution with d degrees of freedom is, *by definition*, the distribution of $Z_1^2 + Z_2^2 + \cdots + Z_d^2$ where the Z_i are *independent* standard normals. So d counts the number of independent standard-normal pieces inside the sum. "Degrees of freedom" literally means "how many independent free coordinates of randomness."

Naive guess vs constraint. We have K approximately-standard-normal z_k, so a first guess is $\chi^2(K)$. But the z_k are not independent. The deviations $d_k = \text{obs}[k] - \text{exp}[k]$ already sum to zero, because both $\sum_k \text{obs}[k]$ and $\sum_k \text{exp}[k]$ equal U. Substituting $d_k = z_k\sqrt{\text{exp}[k]}$ into that constraint gives one weighted linear relation among the z_k:

$$\sum_k z_k \sqrt{\text{exp}[k]} \;=\; 0.$$

(It's not $\sum_k z_k = 0$ because each d_k got divided by a different $\sqrt{\text{exp}[k]}$.) So the z_k live in a $(K-1)$-dimensional subspace – only $K-1$ of them are "free," and the sum of squares is

$$\chi^2_{\text{Pearson}} \;\sim\; \chi^2(K-1) \quad \text{under } H_0.$$

Geometric picture. A point (x,y,z) constrained by $x+y+z=0$ still has three coordinates, but only two free directions; $x^2+y^2+z^2$ is still a well-defined squared length, but its randomness only fills two dimensions. Same story for the z_k.

Why the constraint doesn't break the Pearson formula – the $K=2$ check. The cleanest way to see that the constraint just costs a dof rather than mangling the formula is to take $K=2$ explicitly. Then obs $= (X, U-X)$, expected counts $(Up, U(1-p))$, and $d_1 = X - Up$, $d_2 = -d_1$. Pearson:

$$\chi^2_{\text{Pearson}} \;=\; \frac{(X-Up)^2}{Up} + \frac{(X-Up)^2}{U(1-p)} \;=\; (X-Up)^2\left(\frac{1}{Up} + \frac{1}{U(1-p)}\right) \;=\; \frac{(X-Up)^2}{Up(1-p)}.$$

That last expression is exactly the squared standard binomial z-score, which under the CLT is $\chi^2(1)$ – not $\chi^2(2)$. The two-bin formula collapses to a single squared z-score because the two deviations are negatives of each other; the constraint shows up cleanly as a dof loss. The general K case is the same geometry: the multinomial Gaussian approximation lives on a $(K-1)$-dimensional plane, and Pearson's formula is the squared length on that plane.

Estimating p parameters costs p more dof. If $\hat\theta$ is MLE-fit from the same data, the fitted model is pulled toward the observed counts, shrinking the deviations along p directions in the residual space (one direction per free parameter, by the score equation). This is Wilks' theorem, formalised in §2. The end result:

$$\chi^2_{\text{Pearson}}(\hat\theta) \;\sim\; \chi^2(K-1-p) \quad \text{under } H_0.$$

where K is the number of cells (after Cochran-merging, below) and p is the number of parameters MLE-fit from the data.

What the statistic should look like. Saying "the statistic is $\chi^2(K-1-p)$-distributed" is only useful if we know what numbers $\chi^2(K-1-p)$ typically produces. Two facts about the chi-squared distribution pin it down – both follow directly from the definition $\chi^2(r) = Z_1^2 + \cdots + Z_r^2$, $Z_i \sim N(0,1)$ independent.

Mean. For a single standard normal Z,

$$\mathbb{E}[Z^2] \;=\; \mathrm{Var}(Z) + (\mathbb{E}[Z])^2 \;=\; 1 + 0 \;=\; 1.$$

[1] Sum across r of them and the means add:

$$\mathbb{E}[\chi^2(r)] \;=\; \mathbb{E}[Z_1^2 + \cdots + Z_r^2] \;=\; r.$$

So the expected size of a $\chi^2(r)$ *is just* r.

Spread. **Don't confuse mean and variance here – they are different numbers.** The mean of $\chi^2(r)$ is r (what we just showed). The *variance* is $2r$, not r. By independence of the Z_i, $\mathrm{Var}(\chi^2(r)) = \sum_i \mathrm{Var}(Z_i^2) = 2r$, using the standard-normal fact $\mathrm{Var}(Z^2) = 2$.[2] So the standard deviation is $\sqrt{2r}$ and a typical $\chi^2(r)$ fluctuates by about $\sqrt{2r}$ around its mean r.

Plugging $r = K - 1 - p$:

$$\mathbb{E}[\chi^2_{\text{Pearson}}(\hat{\theta})] \;\approx\; K-1-p, \qquad \text{s.d.} \;\approx\; \sqrt{2(K-1-p)}.$$

This pins down what "small" means. Under H_0 the statistic does *not* land near zero – it lands near $K-1-p$, give or take a few $\sqrt{2(K-1-p)}$. We reject only when the observed value is many $\sqrt{2(K-1-p)}$ away from $K-1-p$, i.e. when the deviations are genuinely bigger than the model's own noise scale. The p-value is the area of the $\chi^2(K-1-p)$ tail above the observed statistic.

1.5 Cochran's rule – when the Gaussian approximation breaks

What Cochran's rule is for. Pearson's statistic is only *approximately* chi-squared. The whole construction uses a chain of approximations – and Cochran's rule polices the weakest link. The chain is:

$$\mathrm{obs}[k] \overset{\text{(i)}}{\approx} \mathrm{Binomial}(U, \pi_k) \overset{\text{(ii)}}{\approx} \mathrm{Poisson}(\exp[k]) \overset{\text{(iii)}}{\approx} N(\exp[k], \exp[k]).$$

Step (i) is the multinomial → per-cell binomial fact. Step (ii) is binomial-to-Poisson, justified when π_k is small and U large (which is why "Poisson" enters the story even though we never modeled the data as Poisson). Step (iii) is the CLT, justified when the Poisson *mean* $\exp[k]$ is large enough that the distribution looks Gaussian. **The chi-squared story depends on (iii) being decent, and (iii) only holds if** $\exp[k]$ **isn't tiny.**

Why the mean matters: Poisson skewness anchors. A Poisson with small mean is sharply right-skewed and has no business being approximated as a symmetric Gaussian:

- $\exp[k] = 1$: mass $\{0.37, 0.37, 0.18, 0.06, \ldots\}$ on $\{0, 1, 2, 3, \ldots\}$ – deeply asymmetric.
- $\exp[k] = 5$: roughly symmetric around the mean, most mass in $[1, 10]$.
- $\exp[k] \geq 10$: Gaussian approximation good enough that the z-score story holds quantitatively.

[1] The identity $\mathbb{E}[Z^2] = \mathrm{Var}(Z) + (\mathbb{E}[Z])^2$ is the variance formula rearranged: $\mathrm{Var}(Z) = \mathbb{E}[(Z-\mu)^2] = \mathbb{E}[Z^2] - 2\mu\mathbb{E}[Z] + \mu^2 = \mathbb{E}[Z^2] - \mu^2$ with $\mu = \mathbb{E}[Z]$, so $\mathbb{E}[Z^2] = \mathrm{Var}(Z) + \mu^2$. Intuitively: "average squared value" = "spread around the mean" + "mean^2 itself".

[2] For $Z \sim N(0,1)$, $\mathrm{Var}(Z^2) = \mathbb{E}[Z^4] - (\mathbb{E}[Z^2])^2 = 3 - 1 = 2$, using the standard-normal fourth moment $\mathbb{E}[Z^4] = 3$.

The rule, the fix. Cochran (1954) showed empirically that the χ^2 approximation to the sampling distribution of Pearson's statistic stays acceptable as long as no cell has $\exp[k] < 1$ and no more than 20% of cells have $\exp[k] < 5$. The widely-used modern rule of thumb tightens this to "$\exp[k] \geq 5$ for every cell" – conservative but easy to apply. **5 is not magical**; it's an anchor for the skewness regime above.

When some bins are too sparse – typically the right tail of a count distribution, where the truncated-Poisson decays exponentially – the fix is to merge adjacent low-exp bins into a single "tail" bin until the constraint is met. The number of bins *after* merging is the K that enters the dof formula $K - 1 - p$.

1.6 From the general framework to Q2(b): two roles of "Poisson"

Everything above is general. Pearson's χ^2 works for any categorical model: $\pi_1, \ldots, \pi_K$ could be fully specified up front (a fair die), parametrised by a Poisson rate λ, a binomial p, a geometric, a normal with unknown mean and variance, etc. The general dof formula is

$$\mathrm{df} \;=\; K - 1 - p,$$

where p is the number of model parameters MLE-fit from the same data: $p = 0$ for a fully-specified model ($\mathrm{df} = K - 1$), $p = 1$ for a Poisson with $\hat{\lambda}$ ($\mathrm{df} = K - 2$), $p = 2$ for a normal with $\hat{\mu}, \hat{\sigma}$ ($\mathrm{df} = K - 3$), and so on.

Two appearances of "Poisson". In Q2(b) "Poisson" shows up in two distinct ways, and it's worth keeping them separate:

- *Noise of a single cell.* The cell count $\mathrm{obs}[k]$ is $\mathrm{Binomial}(U, \pi_k) \approx \mathrm{Poisson}(\exp[k])$ for small π_k. This is the bridge used in Cochran's rule – it says *why* $\exp[k]$ must be large for the Gaussian approximation to hold.
- *The model being tested.* The data are claimed to come from a (truncated) Poisson with one parameter λ: each read-count cell k gets probability $\pi_k(\lambda)$. We MLE-fit $\hat{\lambda}$ from the same data, so $p = 1$ in the dof formula, giving $\mathrm{df} = K - 2$.

The first is a generic noise approximation that justifies the chi-squared machinery in any application; the second is the specific model whose fit we're testing. They're related (both involve a Poisson) but they answer different questions. The general framework was developed without committing to either; the next section formalises what the second one costs in dof.

2 Wilks' correction for estimated parameters

2.1 What changes when the model's parameters are fit from the data

So far we treated $\pi_k(\theta)$ as fixed by an externally-specified θ. In practice we don't know θ – we estimate $\hat{\theta}$ by MLE *from the same data* we're testing. That matters, because $\hat{\theta}$ is chosen specifically to make $\exp[k](\hat{\theta})$ close to $\mathrm{obs}[k]$. The deviations $\mathrm{obs}[k] - \exp[k](\hat{\theta})$ are systematically smaller than $\mathrm{obs}[k] - \exp[k](\theta_0)$ would be at the true parameter, and the Pearson statistic is correspondingly smaller. If we used the un-corrected $\chi^2(K-1)$ null distribution we would essentially never reject the model – the denominator would be tuned to make the numerator look small.

2.2 The geometric picture

Think of the observed histogram as a point in K-dimensional count space.

- The total-count constraint $\sum_k \text{obs}[k] = U$ pins it to a $(K-1)$-dimensional surface.
- A model with no fitted parameters is a single point on that surface (the predicted histogram). Pearson's statistic is the squared normalised distance from obs to that point, and the residual fills $K-1$ directions: $\chi^2(K-1)$.
- A model with p fitted parameters is not a point but a p-dimensional curve / surface (one degree of freedom in $\exp(\theta)$ per parameter). MLE picks the closest point on that curve to obs – it *projects* the observed histogram onto the model surface. The remaining residual lives in the orthogonal complement, which has $K-1-p$ directions.

Each fitted parameter therefore "absorbs" one direction of noise, and the remaining squared distance is $\chi^2(K-1-p)$.

2.3 Wilks' theorem (1938)

Statement. If the model has p free parameters $\theta = (\theta_1, \ldots, \theta_p)$, $\hat{\theta}$ is the MLE from the same data, and the model is correct at some θ_0, then

$$\chi^2_{\text{Pearson}}(\hat{\theta}) \xrightarrow{d} \chi^2(K-1-p) \qquad \text{as } U \to \infty.$$

The total degrees-of-freedom accounting is

$$\text{df} \;=\; (\text{cells after Cochran merging}) - 1 - (\#\text{ parameters MLE-fit from the data}).$$

Reading this as a question. The test now asks: *after giving the model the unfair advantage of picking its best $\hat{\theta}$, are the remaining deviations still too large?* If yes, reject; the data don't fit even when the model is given the benefit of the doubt.

Numerical example. With $K = 10$ bins, a model with fully-specified π_k has df $= 9$. If the same model has one fitted parameter ($\pi_k = \pi_k(\lambda)$ with $\hat{\lambda}$ MLE), df $= 8$, because the fitted λ already absorbs one direction of deviation. For Q2(b) here we estimate λ only, so $p = 1$ and df $= K - 2$.

2.4 Practical effect in Q2(b): why χ^2 is in the tens of thousands

After Cochran tail-merging into $K \approx 9$–13 non-sparse bins, df $= K - 2 \approx 7$–11. Under H_0 we expect $\chi^2 \approx K - 2$ with sd $\sqrt{2(K-2)} \approx 4$. What we actually see on the data is $\chi^2 \sim 10^3$–10^4 – three to four orders of magnitude beyond the null mean. Two compounding factors explain the gap.

(1) Huge U amplifies tiny relative misses. Each cell contributes $z_k^2 = (\text{obs}[k] - \exp[k])^2 / \exp[k]$. Writing $\text{obs}[k] = \exp[k](1 + r_k)$ where r_k is the *relative* miss, this becomes $z_k^2 = \exp[k]\, r_k^2$. So absolute squared deviations scale linearly with $\exp[k]$. With $U \sim 10^6$, a typical bin has $\exp[k] \sim 10^4$–10^5, and a modest 10% relative miss already contributes $z_k^2 \sim 10^2$–10^3 all by itself. A few cells like that and χ^2 hits the thousands.

(2) The right tail catastrophically misfits. The truncated-Poisson predicts an exponentially decaying right tail ($\propto \lambda^k/k!$). The actual histograms have a much heavier right tail because λ is *not* constant across fragments – some amplify a lot ($n_j = 10, 15, 20$), others barely at all ($n_j = 1, 2$). A single $\hat{\lambda}$ averages them and ends up matching neither regime well; in particular it predicts $\exp[k]$ at high k that is orders of magnitude below what's observed. Cochran-merging collapses those into a single "tail" bin, but the merged tail still has obs $\gg$ exp, and that one bin alone can contribute $\sim 10^3$–10^4 to the statistic.

Bottom line. $\chi^2 \sim 10^4$ on df ~ 10 is not borderline – the dof correction is irrelevant. The model is rejected by a margin of hundreds of standard deviations. The *why* – heterogeneous λ across fragments – is exactly what Q3 unpacks (where $\hat{\lambda}_\ell$ is shown to vary by a factor of $\sim$ 5–10 with fragment length alone).

3 Likelihood-ratio test (LRT)

3.1 What an LRT is, in one paragraph

The LRT is the same machinery as Pearson, just framed via likelihoods. You start with a *restricted* model H_0 (lower-dimensional family Θ_0, say p_0 free parameters) and a *fuller* nested model H_1 (higher-dimensional family Θ_1, p_1 free parameters, $p_1 > p_0$). MLE-fit each side; the LRT statistic compares the maximised likelihoods:

$$\Lambda \;=\; \frac{\sup_{\theta\in\Theta_1} L(\theta)}{\sup_{\theta\in\Theta_0} L(\theta)} \;=\; \frac{L(\hat{\theta}_1)}{L(\hat{\theta}_0)}, \qquad T \;:=\; 2\log\Lambda \;\geq\; 0.$$

Because $\Theta_0 \subset \Theta_1$, H_1 can never fit worse than H_0, so $T \geq 0$ always. The question is whether H_1 fits *enough* better to justify rejecting H_0.

3.2 Null distribution (Wilks)

Under H_0, with mild regularity, Wilks gives

$$T \;\xrightarrow{d}\; \chi^2\big(p_1 - p_0\big) \qquad \text{as } U \to \infty,$$

i.e. dof = (free parameters in H_1) − (free parameters in H_0).

Why $p_1 - p_0$ and not p_0 or p_1. The LRT does *not* measure "how noisy H_0 is." It measures *how much better H_1 fits than H_0* – a difference of log-likelihoods. The bigger model can only improve fit by exercising the extra $p_1 - p_0$ parameters that H_0 doesn't have. Under H_0, that improvement is pure overfitting noise filling exactly $p_1 - p_0$ directions, so $T \sim \chi^2(p_1 - p_0)$. The geometry is the same projection-residual story as in §2, just applied to the *gap between two fitted models* instead of *the residual of one fitted model against the data.*

A small example to lock it in. Suppose $H_1 : X \sim N(\mu, \sigma^2)$ ($p_1 = 2$) and $H_0 : X \sim N(0, 1)$ ($p_0 = 0$, both parameters fixed). Then $p_1 - p_0 = 2$ and $T \sim \chi^2(2)$ – even though $p_0 = 0$, so $\chi^2(p_0)$ would be a degenerate point distribution at zero (clearly not what's going on). The two free directions in H_1 (μ and σ^2) can each absorb a unit of noise, giving exactly two-dimensional overfitting under the null.

Pearson vs LRT, same machinery. Both are special cases of the score / Wald / likelihood-ratio asymptotic trinity. They agree under H_0 up to $\mathcal{O}_p(U^{-1})$ in finite samples. The conceptual distinction is what's being projected onto what:

- *Pearson* measures the leftover distance from the *fitted model* to the *data*, so $\mathrm{df} = K - 1 - p$ (cells minus constraint minus fitted parameters).
- *LRT* measures the distance between the *small fitted model* and the *large fitted model*, so $\mathrm{df} = p_1 - p_0$ (extra-parameter count).

We pick whichever is computationally easier: Pearson for Q2(b) (histogram structure), LRT for Q3(b) (rate-parameter nesting across files).

4 Q3(b) LRT in detail

4.1 Exactly what we are testing

Two indices to keep straight. We use two different letters because they play different roles:

- ℓ – a *length bin* (e.g. $\ell = 150\,\mathrm{bp}$). We run a *separate* LRT at each ℓ; ℓ just identifies which one. It stays fixed throughout one test.
- f – a *file index*, $f = 1, 2, \ldots, F$. Within a single test at bin ℓ, we sum over the $F = 4$ files. So f runs inside the sums; ℓ does not.

T_ℓ is the LRT statistic at bin ℓ; $\sum_{f=1}^{F}$ is the sum over files *at* that bin.

Setup. Fix one length bin ℓ. We have F files; file f contributes $(N_{\ell,f}, U_{\ell,f})$. We want to know whether the F files share one PCR rate λ at this bin, or whether each file has its own rate.

H_0 **(shared rate).** A single λ governs all F files: $\lambda_1 = \cdots = \lambda_F = \lambda$. **One free parameter**, $p_0 = 1$.

H_1 **(per-file rate).** Each file gets its own λ_f, no constraint. F **free parameters**, $p_1 = F$.

LRT dof. $p_1 - p_0 = F - 1$. So under H_0, $T_\ell \sim \chi^2(F-1)$ asymptotically. For our four-file case ($F = 4$), $T_\ell \sim \chi^2(3)$.

Common confusion. The "$K - 1 - p$" formula from Pearson does not apply here. It came from a histogram with K cells plus the total-count constraint. The Q3(b) LRT has a different structure – F files (not K cells), no total-count constraint, and the dof is just the extra-parameter count $p_1 - p_0 = F - 1$.

4.2 The log-likelihoods

The per-file truncated-Poisson log-likelihood at rate λ, dropping λ-independent constants, is

$$\log L_f(\lambda) \;=\; -U_{\ell,f}\,\lambda + N_{\ell,f}\log\lambda - U_{\ell,f}\log\bigl(1 - e^{-\lambda}\bigr).$$

4.3 Computing the two MLEs

H_1: fit each file separately. Each file solves Q1's MLE equation $\lambda/(1-e^{-\lambda}) = N_{\ell,f}/U_{\ell,f}$ by `brentq`, giving $\hat{\lambda}_f$. The total H_1 log-likelihood at this bin is $\sum_{f=1}^{F} \log L_f(\hat{\lambda}_f)$.

H_0: fit one λ to the pooled data. Under H_0 all files share λ, so the joint MLE solves the Q1 equation on the pooled sums:[3]

$$\frac{\lambda}{1-e^{-\lambda}} \;=\; \frac{N_{\text{pool}}}{U_{\text{pool}}}, \qquad N_{\text{pool}} \;=\; \sum_f N_{\ell,f}, \qquad U_{\text{pool}} \;=\; \sum_f U_{\ell,f}.$$

This gives a single $\hat{\lambda}_{\text{pool}}$, and the H_0 log-likelihood is $\sum_{f=1}^{F} \log L_f(\hat{\lambda}_{\text{pool}})$ – all F files scored with the same rate.

4.4 The test statistic and decision rule

The statistic.

$$T_\ell \;=\; 2\Big[\sum_{f=1}^{F} \log L_f(\hat{\lambda}_f) \;-\; \sum_{f=1}^{F} \log L_f(\hat{\lambda}_{\text{pool}})\Big] \;\sim\; \chi^2(F-1) \text{ under } H_0.$$

The bracket is H_1's log-likelihood (each file with its own best rate) minus H_0's (each file forced to the pooled rate). $T_\ell \geq 0$ with equality exactly when $\hat{\lambda}_1 = \cdots = \hat{\lambda}_F = \hat{\lambda}_{\text{pool}}$ (the per-file MLEs all happen to coincide with the pooled MLE).

What T_ℓ should look like under H_0. This step mirrors what we did for Pearson in §1.4. Reusing the same chi-squared facts:

$$\mathbb{E}[\chi^2(F-1)] \;=\; F-1, \qquad \text{sd}\big[\chi^2(F-1)\big] \;=\; \sqrt{2(F-1)}.$$

Plugging $F = 4$: under H_0 the statistic T_ℓ should land near

$$\text{mean} \;\approx\; F-1 \;=\; 3, \qquad \text{sd} \;\approx\; \sqrt{2\cdot 3} \;\approx\; 2.4.$$

So T_ℓ in single digits is normal; T_ℓ in the dozens is borderline; T_ℓ in the hundreds means the per-file MLEs disagree by much more than the sampling noise of a shared-rate model allows.

Decision rule from the noise scale. The level-α threshold $\chi^2_{1-\alpha}(F-1)$ is just the cutoff above which only an α-fraction of $\chi^2(F-1)$ realisations land. Pick $\alpha = 0.05$ and $F-1 = 3$: $\chi^2_{0.95}(3) \approx 7.81$, so we reject whenever $T_\ell > 7.81$ (about 3.4 sd's above the null mean). Equivalently, work with the p-value:

$$\boxed{\text{Reject } H_0 \iff T_\ell > \chi^2_{1-\alpha}(F-1) \iff p_\ell \;=\; \Pr\big[\chi^2(F-1) > T_\ell\big] \;<\; \alpha.}$$

p_ℓ is just the upper-tail mass of $\chi^2(F-1)$ beyond the observed T_ℓ – the same translation from a test statistic to a p-value as in Pearson.

[3] *Why does $p_0 = 1$, even though the sufficient statistic per file is the two numbers (N_f, U_f)?* Because Wilks counts *model parameters*, not data dimensions. The model has one rate parameter (λ). The sufficient statistic (N_f, U_f) is a 2-dimensional data summary, but $\hat{\lambda}$ is determined by their *ratio* N_f/U_f through the Q1 equation – one scalar going into the MLE on the parameter side, even though two numbers go in on the data side. For Wilks: parameter count is what matters, so $p_0 = 1$, $p_1 = F$, dof $= F-1$.

4.5 Practical considerations

- **Per-bin sample size.** A bin is included only if $U_{\ell,f} \geq 50$ for at least 2 files. Otherwise the per-file MLE is unreliable and the asymptotic $\chi^2(F-1)$ approximation breaks down.
- **Statistical vs practical significance.** The LRT detects *any* departure from a shared λ. With $N_{\ell,f}$ in the tens of thousands per bin, even a 10% shift between files produces a large T_ℓ. "Statistically significant" here is not the same as "biologically important"; the report flags this.
- **Multiple testing.** With ~ 90 length bins tested independently, the family-wise error rate under no correction is essentially $1 - 0.95^{90} \approx 0.99$. We apply both Bonferroni (α/m_{tests}, FWER, conservative) and Benjamini–Hochberg ($p_{(i)} \leq i\alpha/m_{\text{tests}}$ in sorted order, FDR, more powerful). See Lecture 5.

5 Reading the Q2(b) and Q3(b) results

5.1 Q2(b)

After Cochran-merging, every file gives χ^2 of order 10^3–10^4 on df $\approx$ 7–11, with p-values numerically indistinguishable from 0. *Conclusion:* the global uniform-λ truncated-Poisson model is strongly rejected on every file. The systematic $\hat{M}_{\text{MLE}} > \hat{M}_{\text{fit}}$ gap from Q2(a) is the same misspecification manifesting in the saturation-curve fit.

5.2 Q3(b)

Of ~ 91 length bins, 86 ($\sim 95\%$) reject the shared-λ null at $\alpha = 0.05$ (raw p), 86 ($\sim 95\%$) at BH-FDR $q < 0.05$, and 78 ($\sim 86\%$) at Bonferroni $\alpha/91 \approx 5.5 \times 10^{-4}$. The 78 ($\sim 86\%$) at Bonferroni $\alpha/91 \approx 5.5 \times 10^{-4}$. The p-value distribution is sharply bimodal – bins either reject decisively or not at all – so the three corrections give nearly identical counts. Rejections are dominated by Sample 2 vs the Sample 1 parts (the cross-file amplitude shift visible in the λ vs. length plot).

5.3 Translating back to the science

Both tests confirm that the data are *not* generated by a single truncated-Poisson with one λ. The per-bin LRT then localises where the heterogeneity lives – across files at most lengths, indicating real